## Larkhall NDE Services Inc.

**Inspection Report**

Site: Hamilton Plant

| No. | Status | Date |
|---|---|---|
| LNDE-RPT-20160118-004-DS-6-SG-102 | Issued | 2016-02-12 |

# UT Thickness Data Sheet - 6-SG-102

### Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 6-SG-102, located within the ATU-100. The scope of work is limited to the thickness survey of this specific sour gas inlet line to support the asset integrity assessment.

Marcus Chen, acting as the lead UT technician and SNT-TC-1A UT Level II for Larkhall NDE Services Inc., performed the measurements and is responsible for the accuracy of the recorded data.

### Technical Basis

The inspection was performed under QSCP-NDT-001 Rev 2, which incorporates requirements for insulation removal and reinstatement as part of the CUI program. All thickness measurements were conducted in accordance with ASTM E797.

The data recorded herein is a component of the larger survey detailed in LNDE-RPT-20160118-004-R0, which covers the full-unit survey of the ATU-100.

### Line Specifications

Line 6-SG-102 is constructed from A106 Gr.B with a nominal outer diameter of 168.3 mm and a nominal wall thickness of 7.11 mm. The design pressure is 1965 kPa(g) and the design temperature is 66 °C.

The minimum allowable thickness (t-min) for this line is 3.56 mm. Any reading below this value is subject to an NCR per site rules.

### Analysis and Intervals

Remaining life and corrosion rates are calculated based on the larger of the long-term rate or the short-term rate over the latest pair of readings at least 3 years apart. The next inspection date is determined by half the remaining life or the class maximum.

As a sour gas line, 6-SG-102 is subject to Class 1 thickness and external visual intervals of 5 years.

### Thickness readings, measured 2016-02-12

**Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-SG-101-01 | 7.11 | 3.56 | 2011-02-18 | 6.39 | 6.17 | 0.048 | 0.044 | >50 | 2021-02-12 | |
| 6-SG-101-02 | 7.11 | 3.56 | 2011-02-18 | 6.14 | 5.76 | 0.065 | 0.076 | 28.8 | 2021-02-12 | |
| 6-SG-101-03 | 7.11 | 3.56 | 2011-02-18 | 6.17 | 5.87 | 0.061 | 0.060 | 38.1 | 2021-02-12 | |
| 6-SG-101-04 | 7.11 | 3.56 | 2011-02-18 | 6.05 | 5.58 | 0.076 | 0.094 | 21.4 | 2021-02-12 | |
| 6-SG-101-05 | 7.11 | 3.56 | 2011-02-18 | 5.87 | 5.38 | 0.088 | 0.098 | 18.5 | 2021-02-12 | |
| 6-SG-101-06 | 7.11 | 3.56 | 2011-02-18 | 6.27 | 5.88 | 0.060 | 0.078 | 29.6 | 2021-02-12 | |
| 6-SG-101-08 | 7.11 | 3.56 | 2011-02-18 | 6.33 | 6.00 | 0.066 | 0.066 | 36.8 | 2021-02-12 | |

**Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-SG-102-01 | 7.11 | 3.56 | 2011-02-18 | 5.75 | 5.42 | 0.086 | 0.066 | 21.6 | 2021-02-12 | |
| 6-SG-102-02 | 7.11 | 3.56 | 2011-02-18 | 5.80 | 5.43 | 0.084 | 0.074 | 22.2 | 2021-02-12 | |
| 6-SG-102-03 | 7.11 | 3.56 | 2011-02-18 | 5.80 | 5.22 | 0.091 | 0.116 | 14.3 | 2021-02-12 | |
| 6-SG-102-04 | 7.11 | 3.56 | 2011-02-18 | 5.73 | 5.18 | 0.091 | 0.110 | 14.7 | 2021-02-12 | |
| 6-SG-102-05 | 7.11 | 3.56 | 2011-02-18 | 6.46 | 6.19 | 0.047 | 0.054 | 48.5 | 2021-02-12 | |
| 6-SG-102-06 | 7.11 | 3.56 | 2011-02-18 | 6.41 | 6.06 | 0.070 | 0.070 | 35.6 | 2021-02-12 | |

### Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 8-TG-101-01 | 8.18 | 4.06 | 2006-02-24 | 7.89 | 7.76 | 0.022 | 0.013 | >50 | 2026-02-12 | |
| 8-TG-101-02 | 8.18 | 4.06 | 2006-02-24 | 7.75 | 7.46 | 0.037 | 0.029 | >50 | 2026-02-12 | |
| 8-TG-101-03 | 8.18 | 4.06 | 2006-02-24 | 7.79 | 7.54 | 0.035 | 0.025 | >50 | 2026-02-12 | |
| 8-TG-101-04 | 8.18 | 4.06 | 2006-02-24 | 7.76 | 7.51 | 0.030 | 0.025 | >50 | 2026-02-12 | |
| 8-TG-101-05 | 8.18 | 4.06 | 2006-02-24 | 7.95 | 7.73 | 0.025 | 0.022 | >50 | 2026-02-12 | |
| 8-TG-101-07 | 8.18 | 4.06 | — | — | 7.39 | — | — | — | 2026-02-12 | |

### Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-LA-101-01 | 6.02 | 3.18 | 2006-02-24 | 5.51 | 5.02 | 0.049 | 0.049 | 37.4 | 2026-02-12 | |
| 4-LA-101-02 | 6.02 | 3.18 | 2006-02-24 | 5.76 | 5.45 | 0.028 | 0.031 | >50 | 2026-02-12 | |
| 4-LA-101-03 | 6.02 | 3.18 | 2006-02-24 | 5.76 | 5.45 | 0.026 | 0.031 | >50 | 2026-02-12 | |
| 4-LA-101-04 | 6.02 | 3.18 | 2006-02-24 | 5.78 | 5.48 | 0.030 | 0.030 | >50 | 2026-02-12 | |
| 4-LA-101-05 | 6.02 | 3.18 | 2006-02-24 | 5.81 | 5.54 | 0.024 | 0.027 | >50 | 2026-02-12 | |
| 4-LA-101-06 | 6.02 | 3.18 | 2006-02-24 | 5.64 | 5.30 | 0.038 | 0.034 | >50 | 2026-02-12 | |
| 4-LA-101-07 | 6.02 | 3.18 | 2006-02-24 | 5.57 | 4.87 | 0.056 | 0.070 | 24.1 | 2026-02-12 | |
| 4-LA-101-09 | 6.02 | 3.18 | — | — | 5.00 | — | — | — | 2026-02-12 | |

### Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-LA-102-01 | 6.02 | 3.18 | 2006-02-24 | 5.63 | 5.25 | 0.039 | 0.038 | >50 | 2026-02-12 | |
| 4-LA-102-02 | 6.02 | 3.18 | 2006-02-24 | 5.67 | 5.36 | 0.036 | 0.031 | >50 | 2026-02-12 | |
| 4-LA-102-03 | 6.02 | 3.18 | 2006-02-24 | 5.55 | 5.28 | 0.040 | 0.027 | >50 | 2026-02-12 | |
| 4-LA-102-04 | 6.02 | 3.18 | 2006-02-24 | 5.34 | 4.90 | 0.055 | 0.044 | 31.5 | 2026-02-12 | |
| 4-LA-102-05 | 6.02 | 3.18 | 2006-02-24 | 5.62 | 5.34 | 0.033 | 0.028 | >50 | 2026-02-12 | |
| 4-LA-102-06 | 6.02 | 3.18 | 2006-02-24 | 5.83 | 5.60 | 0.020 | 0.023 | >50 | 2026-02-12 | |
| 4-LA-102-07 | 6.02 | 3.18 | 2006-02-24 | 5.63 | 5.40 | 0.030 | 0.023 | >50 | 2026-02-12 | |
| 4-LA-102-08 | 6.02 | 3.18 | — | — | 5.70 | — | — | — | 2026-02-12 | |

### Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-LA-103-01 | 7.11 | 3.56 | 2006-02-24 | 6.75 | 6.44 | 0.029 | 0.031 | >50 | 2026-02-12 | |
| 6-LA-103-02 | 7.11 | 3.56 | 2006-02-24 | 6.87 | 6.53 | 0.033 | 0.034 | >50 | 2026-02-12 | |
| 6-LA-103-03 | 7.11 | 3.56 | 2006-02-24 | 6.65 | 6.28 | 0.040 | 0.037 | >50 | 2026-02-12 | |
| 6-LA-103-04 | 7.11 | 3.56 | 2006-02-24 | 6.52 | 6.15 | 0.051 | 0.037 | >50 | 2026-02-12 | |
| 6-LA-103-05 | 7.11 | 3.56 | 2006-02-24 | 6.49 | 5.98 | 0.061 | 0.051 | 39.9 | 2026-02-12 | |
| 6-LA-103-06 | 7.11 | 3.56 | 2006-02-24 | 6.42 | 5.82 | 0.067 | 0.060 | 33.6 | 2026-02-12 | |
| 6-LA-103-07 | 7.11 | 3.56 | 2006-02-24 | 6.76 | 6.12 | 0.053 | 0.064 | 39.9 | 2026-02-12 | |
| 6-LA-103-08 | 7.11 | 3.56 | 2006-02-24 | 6.79 | 6.49 | 0.028 | 0.030 | >50 | 2026-02-12 | |
| 6-LA-103-10 | 7.11 | 3.56 | — | — | 6.30 | — | — | — | 2026-02-12 | |

### Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-101-01 | 8.56 | 3.81 | 2014-09-19 | 8.07 | 7.83 | 0.035 | 0.098 | 40.9 | 2019-02-12 | |
| 4-RA-101-02 | 8.56 | 3.81 | 2014-09-19 | 8.02 | 7.94 | 0.030 | 0.052 | >50 | 2019-02-12 | |
| 4-RA-101-03 | 8.56 | 3.81 | 2014-09-19 | 7.88 | 7.65 | 0.048 | 0.066 | >50 | 2019-02-12 | |
| 4-RA-101-04 | 8.56 | 3.81 | 2014-09-19 | 8.18 | 8.00 | 0.032 | 0.042 | >50 | 2019-02-12 | |
| 4-RA-101-05 | 8.56 | 3.81 | 2014-09-19 | 7.98 | 7.82 | 0.039 | 0.068 | >50 | 2019-02-12 | |
| 4-RA-101-06 | 8.56 | 3.81 | 2014-09-19 | 7.91 | 7.87 | 0.029 | 0.029 | >50 | 2019-02-12 | |

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-101-07 | 8.56 | 3.81 | 2014-09-19 | 8.22 | 8.14 | 0.023 | 0.032 | >50 | 2019-02-12 | |
| 4-RA-101-08 | 8.56 | 3.81 | 2014-09-19 | 7.71 | 7.31 | 0.069 | 0.142 | 24.6 | 2019-02-12 | |
| 4-RA-101-09 | 8.56 | 3.81 | 2014-09-19 | 8.27 | 8.13 | 0.022 | 0.038 | >50 | 2019-02-12 | |
| 4-RA-101-10 | 8.56 | 3.81 | 2014-09-19 | 8.21 | 8.02 | 0.033 | 0.052 | >50 | 2019-02-12 | |
| 4-RA-101-11 | 8.56 | 3.81 | 2014-09-19 | 7.88 | 7.60 | 0.054 | 0.088 | 42.9 | 2019-02-12 | |

**Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-102-01 | 8.56 | 3.81 | 2014-09-19 | 8.25 | 8.02 | 0.026 | 0.062 | >50 | 2019-02-12 | |
| 4-RA-102-02 | 8.56 | 3.81 | 2014-09-19 | 7.89 | 7.64 | 0.047 | 0.070 | >50 | 2019-02-12 | |
| 4-RA-102-03 | 8.56 | 3.81 | 2014-09-19 | 7.95 | 7.55 | 0.051 | 0.098 | 38.0 | 2019-02-12 | |
| 4-RA-102-04 | 8.56 | 3.81 | 2014-09-19 | 7.93 | 7.72 | 0.041 | 0.048 | >50 | 2019-02-12 | |
| 4-RA-102-05 | 8.56 | 3.81 | 2014-09-19 | 7.84 | 7.67 | 0.044 | 0.080 | 48.1 | 2019-02-12 | |
| 4-RA-102-06 | 8.56 | 3.81 | 2014-09-19 | 7.95 | 7.73 | 0.039 | 0.056 | >50 | 2019-02-12 | |
| 4-RA-102-07 | 8.56 | 3.81 | 2014-09-19 | 7.98 | 7.78 | 0.040 | 0.064 | >50 | 2019-02-12 | |
| 4-RA-102-08 | 8.56 | 3.81 | 2014-09-19 | 7.58 | 7.35 | 0.057 | 0.106 | 33.3 | 2019-02-12 | |
| 4-RA-102-09 | 8.56 | 3.81 | 2014-09-19 | 7.84 | 7.54 | 0.051 | 0.088 | 42.2 | 2019-02-12 | |
| 4-RA-102-11 | 8.56 | 3.81 | 2014-09-19 | 8.12 | 7.96 | 0.046 | 0.046 | >50 | 2019-02-12 | |

**Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-RA-103-01 | 10.97 | 4.57 | 2014-09-19 | 10.62 | 10.29 | 0.036 | 0.036 | >50 | 2019-02-12 | |
| 6-RA-103-02 | 10.97 | 4.57 | 2014-09-19 | 10.06 | 9.69 | 0.070 | 0.108 | 47.2 | 2019-02-12 | |
| 6-RA-103-03 | 10.97 | 4.57 | 2014-09-19 | 10.39 | 10.06 | 0.047 | 0.072 | >50 | 2019-02-12 | |
| 6-RA-103-04 | 10.97 | 4.57 | 2014-09-19 | 9.79 | 9.31 | 0.084 | 0.132 | 35.8 | 2019-02-12 | |
| 6-RA-103-05 | 10.97 | 4.57 | 2014-09-19 | 9.81 | 9.48 | 0.074 | 0.124 | 39.5 | 2019-02-12 | |
| 6-RA-103-06 | 10.97 | 4.57 | 2014-09-19 | 10.43 | 10.32 | 0.035 | 0.048 | >50 | 2019-02-12 | |
| 6-RA-103-07 | 10.97 | 4.57 | 2014-09-19 | 10.37 | 10.20 | 0.040 | 0.070 | >50 | 2019-02-12 | |
| 6-RA-103-08 | 10.97 | 4.57 | 2014-09-19 | 10.49 | 10.33 | 0.032 | 0.022 | >50 | 2019-02-12 | |
| 6-RA-103-09 | 10.97 | 4.57 | 2014-09-19 | 10.39 | 10.33 | 0.034 | 0.046 | >50 | 2019-02-12 | |
| 6-RA-103-10 | 10.97 | 4.57 | 2014-09-19 | 9.93 | 9.49 | 0.077 | 0.112 | 43.8 | 2019-02-12 | |
| 6-RA-103-11 | 10.97 | 4.57 | 2014-09-19 | 10.10 | 9.77 | 0.236 | 0.236 | 22.0 | 2019-02-12 | |
| 6-RA-103-13 | 10.97 | 4.57 | 2014-09-19 | 10.38 | 10.15 | 0.060 | 0.060 | >50 | 2019-02-12 | |

**Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-RV-101-01 | 7.62 | 3.56 | 2013-09-09 | 6.49 | 6.34 | 0.063 | 0.081 | 34.2 | 2019-02-12 | |
| 3-RV-101-02 | 7.62 | 3.56 | 2013-09-09 | 6.44 | 6.31 | 0.063 | 0.052 | 43.5 | 2019-02-12 | |
| 3-RV-101-03 | 7.62 | 3.56 | 2013-09-09 | 5.80 | 5.63 | 0.095 | 0.098 | 21.2 | 2019-02-12 | |
| 3-RV-101-04 | 7.62 | 3.56 | 2013-09-09 | 6.13 | 5.88 | 0.086 | 0.102 | 22.8 | 2019-02-12 | |
| 3-RV-101-05 | 7.62 | 3.56 | 2013-09-09 | 6.48 | 6.36 | 0.066 | 0.050 | 42.6 | 2019-02-12 | |
| 3-RV-101-06 | 7.62 | 3.56 | 2013-09-09 | 6.11 | 5.96 | 0.082 | 0.087 | 27.7 | 2019-02-12 | |
| 3-RV-101-07 | 7.62 | 3.56 | 2013-09-09 | 5.73 | 5.54 | 0.122 | 0.122 | 16.2 | 2019-02-12 | |
| 3-RV-101-08 | 7.62 | 3.56 | 2013-09-09 | 6.36 | 6.31 | 0.074 | 0.074 | 37.2 | 2019-02-12 | |

**Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-RV-102-01 | 5.54 | 2.79 | 2013-09-09 | 4.49 | 4.35 | 0.059 | 0.037 | 26.4 | 2019-02-12 | |
| 2-RV-102-02 | 5.54 | 2.79 | 2013-09-09 | 4.29 | 4.01 | 0.079 | 0.109 | 11.2 | 2019-02-12 | |
| 2-RV-102-03 | 5.54 | 2.79 | 2013-09-09 | 4.68 | 4.64 | 0.042 | 0.037 | 43.9 | 2019-02-12 | |

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-RV-102-04 | 5.54 | 2.79 | 2013-09-09 | 4.81 | 4.73 | 0.043 | 0.020 | 45.5 | 2019-02-12 | |
| 2-RV-102-05 | 5.54 | 2.79 | 2013-09-09 | 4.96 | 4.97 | 0.013 | 0.013 | >50 | 2019-02-12 | |
| 2-RV-102-06 | 5.54 | 2.79 | 2013-09-09 | 4.55 | 4.27 | 0.102 | 0.102 | 14.6 | 2019-02-12 | |

**Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-RF-101-01 | 5.49 | 2.41 | 2013-12-09 | 5.37 | 5.31 | 0.009 | 0.017 | >50 | 2026-02-12 | |
| 3-RF-101-02 | 5.49 | 2.41 | 2013-12-09 | 5.40 | 5.34 | 0.006 | 0.003 | >50 | 2026-02-12 | |
| 3-RF-101-03 | 5.49 | 2.41 | 2013-12-09 | 5.37 | 5.33 | 0.006 | 0.012 | >50 | 2026-02-12 | |
| 3-RF-101-04 | 5.49 | 2.41 | 2013-12-09 | 5.41 | 5.35 | 0.006 | -0.002 | >50 | 2026-02-12 | |
| 3-RF-101-05 | 5.49 | 2.41 | 2013-12-09 | 5.32 | 5.33 | 0.006 | 0.007 | >50 | 2026-02-12 | |
| 3-RF-101-06 | 5.49 | 2.41 | 2013-12-09 | 5.17 | 5.23 | 0.015 | 0.017 | >50 | 2026-02-12 | |
| 3-RF-101-07 | 5.49 | 2.41 | 2013-12-09 | 5.48 | 5.37 | 0.003 | -0.002 | >50 | 2026-02-12 | |
| 3-RF-101-09 | 5.49 | 2.41 | 2013-12-09 | 5.26 | 5.19 | -0.029 | -0.029 | >50 | 2026-02-12 | |

**Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-RF-102-01 | 3.91 | 2.16 | 2013-12-09 | 3.60 | 3.76 | 0.008 | 0.009 | >50 | 2026-02-12 | |
| 2-RF-102-02 | 3.91 | 2.16 | 2013-12-09 | 3.70 | 3.64 | 0.012 | 0.024 | >50 | 2026-02-12 | |
| 2-RF-102-03 | 3.91 | 2.16 | 2013-12-09 | 3.79 | 3.75 | 0.009 | 0.005 | >50 | 2026-02-12 | |
| 2-RF-102-04 | 3.91 | 2.16 | 2013-12-09 | 3.82 | 3.38 | 0.025 | 0.043 | 28.3 | 2026-02-12 | |
| 2-RF-102-05 | 3.91 | 2.16 | 2013-12-09 | 3.67 | 3.49 | 0.082 | 0.082 | 16.1 | 2024-03-06 | |
| 2-RF-102-06 | 3.91 | 2.16 | 2013-12-09 | 3.67 | 3.47 | 0.099 | 0.099 | 13.2 | 2022-09-26 | |
| 2-RF-102-07 | 3.91 | 2.16 | 2013-12-09 | 3.72 | 3.57 | 0.020 | 0.028 | >50 | 2026-02-12 | |

**Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-LA-104-01 | 6.02 | 3.18 | 2006-02-24 | 5.75 | 5.31 | 0.038 | 0.044 | 48.2 | 2026-02-12 | |
| 4-LA-104-02 | 6.02 | 3.18 | 2006-02-24 | 5.56 | 5.00 | 0.052 | 0.056 | 32.4 | 2026-02-12 | |
| 4-LA-104-03 | 6.02 | 3.18 | 2006-02-24 | 5.74 | 5.42 | 0.027 | 0.032 | >50 | 2026-02-12 | |
| 4-LA-104-04 | 6.02 | 3.18 | 2006-02-24 | 5.86 | 5.63 | 0.022 | 0.023 | >50 | 2026-02-12 | |
| 4-LA-104-05 | 6.02 | 3.18 | 2006-02-24 | 5.51 | 4.88 | 0.058 | 0.063 | 26.9 | 2026-02-12 | |
| 4-LA-104-06 | 6.02 | 3.18 | 2006-02-24 | 5.83 | 5.55 | 0.028 | 0.028 | >50 | 2026-02-12 | |
| 4-LA-104-07 | 6.02 | 3.18 | 2006-02-24 | 5.62 | 5.28 | 0.037 | 0.034 | >50 | 2026-02-12 | |
| 4-LA-104-08 | 6.02 | 3.18 | 2006-02-24 | 5.50 | 5.11 | 0.042 | 0.039 | 46.3 | 2026-02-12 | |
| 4-LA-104-10 | 6.02 | 3.18 | — | — | 4.84 | — | — | — | 2026-02-12 | |

**Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-104-01 | 8.56 | 3.81 | 2014-09-19 | 7.84 | 7.71 | 0.040 | 0.048 | >50 | 2019-02-12 | |
| 4-RA-104-02 | 8.56 | 3.81 | 2014-09-19 | 8.09 | 7.85 | 0.035 | 0.058 | >50 | 2019-02-12 | |
| 4-RA-104-03 | 8.56 | 3.81 | 2014-09-19 | 8.46 | 8.28 | 0.057 | 0.056 | >50 | 2019-02-12 | |
| 4-RA-104-04 | 8.56 | 3.81 | 2014-09-19 | 8.49 | 8.18 | 0.063 | 0.084 | >50 | 2019-02-12 | |
| 4-RA-104-05 | 8.56 | 3.81 | 2014-09-19 | 8.37 | 8.16 | 0.050 | 0.082 | >50 | 2019-02-12 | |
| 4-RA-104-06 | 8.56 | 3.81 | 2014-09-19 | 7.91 | 7.61 | 0.049 | 0.074 | >50 | 2019-02-12 | |
| 4-RA-104-07 | 8.56 | 3.81 | 2014-09-19 | 7.61 | 7.32 | 0.064 | 0.092 | 38.0 | 2019-02-12 | |
| 4-RA-104-08 | 8.56 | 3.81 | 2014-09-19 | 8.00 | 7.81 | 0.074 | 0.074 | >50 | 2019-02-12 | |

**Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-ST-101-01 | 3.91 | 2.29 | 2006-02-24 | 3.64 | 3.61 | 0.016 | 0.003 | >50 | 2026-02-12 | |
| 2-ST-101-02 | 3.91 | 2.29 | 2006-02-24 | 3.67 | 3.55 | 0.018 | 0.012 | >50 | 2026-02-12 | |
| 2-ST-101-03 | 3.91 | 2.29 | 2006-02-24 | 3.82 | 3.55 | 0.020 | 0.027 | 46.5 | 2026-02-12 | |
| 2-ST-101-04 | 3.91 | 2.29 | 2006-02-24 | 3.77 | 3.68 | 0.007 | 0.009 | >50 | 2026-02-12 | |
| 2-ST-101-05 | 3.91 | 2.29 | 2006-02-24 | 3.72 | 3.50 | 0.021 | 0.022 | >50 | 2026-02-12 | |
| 2-ST-101-06 | 3.91 | 2.29 | — | — | 3.63 | — | — | — | 2026-02-12 | |

**Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-CW-101-01 | 5.49 | 2.79 | 2006-02-24 | 5.13 | 5.03 | 0.020 | 0.010 | >50 | 2026-02-12 | |
| 3-CW-101-02 | 5.49 | 2.79 | 2006-02-24 | 5.20 | 5.01 | 0.020 | 0.019 | >50 | 2026-02-12 | |
| 3-CW-101-03 | 5.49 | 2.79 | 2006-02-24 | 5.08 | 4.93 | 0.030 | 0.015 | >50 | 2026-02-12 | |
| 3-CW-101-04 | 5.49 | 2.79 | 2006-02-24 | 5.38 | 5.26 | 0.010 | 0.012 | >50 | 2026-02-12 | |
| 3-CW-101-05 | 5.49 | 2.79 | 2006-02-24 | 5.33 | 5.35 | 0.007 | -0.002 | >50 | 2026-02-12 | |
| 3-CW-101-06 | 5.49 | 2.79 | — | — | 5.08 | — | — | — | 2026-02-12 | |

**Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-CW-102-01 | 5.49 | 2.79 | 2006-02-24 | 5.20 | 5.06 | 0.022 | 0.014 | >50 | 2026-02-12 | |
| 3-CW-102-02 | 5.49 | 2.79 | 2006-02-24 | 5.02 | 4.37 | 0.057 | 0.065 | 24.2 | 2026-02-12 | |
| 3-CW-102-03 | 5.49 | 2.79 | 2006-02-24 | 5.07 | 4.54 | 0.050 | 0.053 | 32.9 | 2026-02-12 | |
| 3-CW-102-04 | 5.49 | 2.79 | 2006-02-24 | 5.03 | 4.69 | 0.041 | 0.034 | 46.2 | 2026-02-12 | |
| 3-CW-102-05 | 5.49 | 2.79 | — | — | 4.66 | — | — | — | 2026-02-12 | |

| Sign-off | Name | Function | Date | Signature |
|---|---|---|---|---|
| Prepared | Marcus Chen | lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II | 2016-02-12 | _____ |